#ManyDaughters: Pre-Analysis Plan

RT

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1 Introduction

Our analysis focuses on gender equality in intra-household time allocation. This topic has been studied in economics at least since Gronau (1973). While time allocation between genders has become more equal over time, women still do more unpaid work, while men do more paid work on average (Gimenez-Nadal and Sevilla 2012). We will explore what effect having daughters has on attitudes and behaviors around time allocation. This research question is not totally unexplored, as there exists prior literature on this issue (Lundberg and Rose 2002; Bhalotra et al. 2025). One article even uses the SOEP dataset to explore child gender and the division of labor (Pollmann-Schult 2017). However, the data ends in 2011 and we will use specific questions added to the survey after this date. Besides, for this type of meta-analysis, the novelty of the research question is of second order importance.

2 Variables

2.1 Outcome variables

For each hypothesis, we specify one dependent variable that we will use to test the hypothesis.

H1 (attitude): Agreement to the statement *"It's best if the man and the woman work the same amount so they can share the responsibility for taking care of the family and household equally"* [SOEP Variable code plh0308_v2].

Original text (German): *"Am besten ist es, wenn der Mann und die Frau beide gleich viel erwerbstätig sind und sich beide in gleichem Maße um Haushalt und Familie kümmern."* This variable is part of the 2018 survey wave and asked to the core sample of SOEP participants (initial sample and refreshment samples). As such, the variable is available for one cross-section (and not a panel).[1] Therefore, our analysis is based on between-subject variation. The question asks for agreement on a seven-point Likert-scale ranging from 1 (disagree completely) to 7 (agree completely). As a higher value indicates a stronger support for gender equality, it can be used without recoding for "X" as defined in #ManyDaughters Information for RTs.

H2 (political preference): Satisfaction rating based on question *"How satisfied are you with the level of social justice in Germany?"* [SOEP Variable code plh0355]

Original text (German): *"Wie zufrieden sind Sie mit der Verwirklichung der sozialen Gerechtigkeit in Deutschland?"* This variable is part of the 2015 survey wave and asked to the core sample of SOEP participants (initial sample and refreshment samples). As such, the variable is available for one cross-section (and not a panel). Therefore, our analysis is based on between-subject variation. The question asks for a satisfaction rating on an eleven-point scale ranging from 0 (completely dissatisfied) to 10 (completely satisfied). A lower value on this scale indicates more left leaning political preferences, as it suggests a preference for a higher level of social justice. We thus multiply the variable by (-1) to use it for "Y" as defined in #ManyDaughters Information for RTs.

H3 (behavior): Equality in housework between partners based on self-reported hours spent on housework "How many hours do you spend on the following activities on a typical weekday, Saturday, and Sunday?" Housework (washing, cooking, cleaning) [SOEP Variable codes pli0012_h, pli0016_h, pli0043_h]

Original text (German): *"Wie viele Stunden pro Tag entfallen bei Ihnen an einem durchschnittlichen Werktag, an einem typischen Samstag und an einem typischen Sonntag auf die folgenden Tätigkeiten? Hausarbeit (Waschen, Kochen, Putzen)"*. This variable

[1] We do not use the related variable plh0308_v1 as it is asked in a different, non-representative sample with fewer observations and older data. In addition, it is elicited on a different scale.

is included in multiple survey waves and repeatedly asked to the core sample of SOEP participants. The questions ask for whole hours allocated per day. We aggregate the weekday and weekend hours to total hours spent on housework per week. We identify the partner of the respondent and calculate the difference in total hours worked per week. We scale the absolute difference by total hours worked between the couple and reverse code it to obtain a gender equality variable to use it for "Z" as defined in #ManyDaughters Information for RTs. The resulting variable is between 0 and 1 with a value of 1 if both partners work the same hours and 0 if one partner does all the housework.

H4 (attitude x gender): We use the agreement to sharing responsibilities as stated in H1 also for the evaluation of H4.

As H4 in #ManyDaughters is defined, it suggests to reuse the "X" variable from H1. We follow this instruction.

2.2 Main explanatory variable

In all regressions (H1-H4), we use the number of daughters as independent variable of interest. In prior research, the number of daughters has been used for example by Washington (2008) and Oswald and Powdthavee (2010). As only one regression specification can be tested for each hypothesis, other plausible variable definitions such as a binary daughter variable remain untested. The number of daughters is a count variable that takes a value of 0 for participants with no children or only sons. As H3 is estimated on household level, we use the maximum number of daughters between partners in case the number of daughters differs within a couple. For H4, we interact the number of daughters with the gender of the respondent.

Remark on causality: As in much of prior research, we treat the gender of offspring as exogenous, supporting a causal interpretation of the results. Selective abortion should be a minor issue in Germany, as the gender of an embryo may only be revealed after the legal abortion limit. Family stopping rules usually also have a minor effect on the number of daughters, if couples, for example, continue having children until they have a son. Furthermore, the choice to have children at all clearly is an endogenous decision. We control for both issues by including number of children fixed effects (see control variables). We decide to keep participants without children in the sample, as in #ManyDaughters hypotheses are defined

relative to respondents "without daughters", not relative to respondents with only sons.

2.3 Control variables

We use the following control variables:

- Number of children: Set of dummy variables for number of children.
- Age: Age in years using the survey year minus the birth year [syear-gebjahr].
- Female: Indicator variable (0/1) based on indicated sex [sex].
- Education: Years of education [pgbilzeit].
- Log income: Natural logarithm of monthly household income [ln(hghinc)].
- Married: Indicator variable (0/1) married or registered civil partnership [pgfamstd].
- German: Indicator variable (0/1) German citizen [plj0014_v3].

For H3 the control variables are adjusted to account for the household (couple) level:

- Number of children: Set of indicator variables for maximum number of children between partners.
- Age: Mean age in years between respondent and partner.
- Female: Not used on household level (one observation represents a mixed-sex couple.
- Education: Mean education in years between respondent and partner.
- Log income: As before.
- Married: As before.
- German: Share of German citizens in a couple (0/0.5/1).

As H3 is estimated in a panel, we additionally use year fixed-effects based on survey year [syear].

3 Inclusions and exclusions

Hypothesis H1, H2, and H4 are tested on the individual level. Therefore we include all adult respondents (age≥18), who answer the respective questions on attitudes and political preferences. We recode negative values to missing, as in the SOEP various negative numbers indicate that the question was not part of the questionnaire, not included in a particular version, or not answered.

Hypothesis H3 is tested on household level and requires a couple of mixed gender to live within one household. We thus exclude couples that live in separate households. We also exclude same sex couples as for them the time allocation lacks a gender equality component. We exclude couples were one partner is married to a different person. We further exclude singles without a partner and respondents that cannot be matched by partner id [parid].

For H3, we exclude couples for which total hours of housework is zero.[2] We also exclude respondents with >100 hours of housework per week (implausible values). Finally, we only keep one observation per household and survey year for the analysis.

4 Regression models and interpretation

To test hypothesis **H1**, we use the following linear regression model:

$$\begin{aligned} Share_responsibilty = \beta_0 + \beta_1 \cdot num_daughters + \beta_2 \cdot female + \beta_3 \cdot age + \beta_4 \cdot education \\ + \beta_5 \cdot log_income + \beta_6 \cdot married + \beta_7 \cdot german + \Gamma \cdot \mathrm{I}_{num_children} + \varepsilon \end{aligned}$$

The regression is estimated using OLS and clustered standard errors on household level [reg with vce(cluster hid)]. The relevant coefficient to test H1 is β_1 which is expected to be positive if H1 holds (having daughters increases agreement that responsibilities at work and in the household should be shared more equally). We use the t-statistic of this coefficient and its associated p-value as relevant criterion for statistical significance. In line with Benjamin et al. (2018), we interpret $p < 0.05$ as suggestive evidence and $p < 0.005$ as significant evidence to reject the null hypothesis of no effect.

[2] This suggests that another person in the household or paid external person does the entire housework.

To test hypothesis **H2**, we use the following linear regression model:

$$\begin{aligned} Satisfaction_social = \beta_0 + \beta_1 \cdot num_daughters + \beta_2 \cdot female + \beta_3 \cdot age + \beta_4 \cdot education \\ + \beta_5 \cdot log_income + \beta_6 \cdot married + \beta_7 \cdot german + \Gamma \cdot \mathrm{I}_{num_children} + \varepsilon \end{aligned}$$

The regression is estimated using OLS and clustered standard errors on household level [reg with vce(cluster hid)]. The relevant coefficient to test H2 is β_1 which is expected to be positive if H2 holds (having daughters decreases satisfaction with social justice in Germany [inversely coded].). We use the t-statistic of this coefficient and its associated p-value as relevant criterion for statistical significance. In line with Benjamin et al. (2018), we interpret $p < 0.05$ as suggestive evidence and $p < 0.005$ as significant evidence to reject the null hypothesis of no effect.

To test hypothesis **H3**, we use the following linear regression model:

$$\begin{aligned} Housework_equality = \beta_0 + \beta_1 \cdot max_daughters_{hh} + \beta_2 \cdot mean_age_{hh} + \beta_3 \cdot mean_education_{hh} \\ + \beta_4 \cdot log_income + \beta_5 \cdot married + \beta_6 \cdot german_share_{hh} \\ + \Gamma \cdot \mathrm{I}_{num_children} + \Phi \cdot \mathrm{I}_{survey_year} + \varepsilon \end{aligned}$$

The regression is estimated using a pooled panel regression (OLS) and clustered standard errors on household level. The regression includes survey year fixed effects, as the intra-household distribution of housework has presumably changed over time. The relevant coefficient to test H3 is β_1 which is expected to be positive if H3 holds (having daughters as a couple increases gender equality in the allocation of housework). We use the t-statistic of this coefficient and its associated p-value as relevant criterion for statistical significance. In line with Benjamin et al. (2018), we interpret $p < 0.05$ as suggestive evidence and $p < 0.005$ as significant evidence to reject the null hypothesis of no effect.

To test hypothesis **H4**, we use the following linear regression model:

$$\begin{aligned} Share_responsibilty = \beta_0 + \beta_1 \cdot (num_daughters \times female) + \beta_2 \cdot num_daughters \\ + \beta_3 \cdot female + \beta_4 \cdot age + \beta_5 \cdot education + \beta_6 \cdot log_income \\ + \beta_7 \cdot married + \beta_8 \cdot german + \Gamma \cdot \mathrm{I}_{num_children} + \varepsilon \end{aligned}$$

The regression is estimated using OLS and clustered standard errors on household level [reg with vce(cluster hid)]. It includes an interaction term between the number of daughters and the female indicator. The relevant coefficient to test H4 is β_1, which is expected to be negative if H4 holds (female respondents with daughters increase their agreement to shared responsibilities less than male respondents). We use the t-statistic of this coefficient and its associated p-value as relevant criterion for statistical significance. In line with Benjamin et al. (2018), we interpret $p < 0.05$ as suggestive evidence and $p < 0.005$ as significant evidence to reject the null hypothesis of no effect.

References

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